

Md Arafat Alam Khandaker

✉ aarafatalam18@gmail.com in Md Arafat Alam Khandaker 📧 AArafatt 🌐 AArafatt.github.io

Research Interest

Machine Learning and its applications, Deep Learning, Computer Vision, Natural Language Processing, Medical Image Analysis, Large Language Models, Audio Data Analysis.

Education

Ahsanullah University of Science and Technology
BSc in Computer Science and Engineering

Dec 2020 – January 2025

- CGPA: 3.04/4.0
- **Undergrad Thesis Title:** Multimodal Approaches for Software Issue Classification: Leveraging CLIP and Advanced Fusion Techniques **Published** [🔗](#)
- Supervisor:** Dr. Md. Shamim Akhter, Professor, Department of CSE, Ahsanullah University of Science and Technology [🔗](#)

Work Experince

Software Engineering (AI/ML)
Cloudly Infotech Limited

June 2025 – Present

- Designed an AI-powered HR Intelligence Agent to automate HR operations and provide workforce insights via natural language interaction with company databases
- Cloudly Orchestrator – AI-driven network-log analysis platform that parses multi-vendor logs, auto-detects roles, and delivers real-time troubleshooting, security and compliance insights via LangChain & LLMs

Experience

Research Assistant

Applied Intelligent System and Information Processing Lab (AISIP)

Sep 2023 – Present

Supervisor: Dr. Md. Shamim Akhter, Research Professor, Ahsanullah University of Science and Technology

- Carried out research on **"Multimodal Classification of Software Issue Reports Using CLIP: A Fine-Tuned Approach for Bug and Feature Detection"**.
- Conducted research on **"Navigating High-Dimensional Data with Advanced Clustering Algorithms"**
- Currently working on the research titled **"SarcFormer: A Transformer Based Framework for Sarcasm Analysis"**

Publications

[Google Scholar](#) [🔗](#)

Conference Proceedings.....

- **Md Arafat Alam Khandaker**, Ziyen Shirin Raha, Shifat Islam and Tashreef Muhammad. "Explainable AI-Enhanced Deep Learning for Pumpkin Leaf Disease Detection: A Comparative Analysis of CNN Architectures." [Paper link](#) [🔗](#)
- **Md Arafat Alam Khandaker**, Ziyen Shirin Raha, Salehin Bin Iqbal, M.F. Mridha and Jungpil Shin. "From Images to Insights: Transforming Brain Cancer Diagnosis with Explainable AI." [Paper link](#) [🔗](#)
- **Md Arafat Alam Khandaker**, Ziyen Shirin Raha, Bidyarthi Paul and Tashreef Muhammad. "Bridging Dialects: Translating Standard Bangla to Regional Variants Using Neural Models." [Paper link](#) [🔗](#)
- Ziyen Shirin Raha, **Md Arafat Alam Khandaker**, Md. Zahid Hossain and Shamim Akhter. "Navigating High-Dimensional Data with Advanced Clustering Algorithms." [Paper link](#) [🔗](#)

- Yasir Arafah Prince, Sumaiya Shejin, Shamim Akhter, **Md Arafat Alam Khandaker** “Multimodal Classification of Software Issue Reports Using CLIP: A Fine-Tuned Approach for Bug and Feature Detection.” [Paper link](#) 

Journals.....

- Fatema Tuj Johora Faria, Mukaffi Bin Moin, Zayeed Hasan, **Md Arafat Alam Khandaker**, Niful Islam, Khan Md Hasib, and M. F. Mridha. “**MultiBanFakeDetect: Integrating Advanced Fusion Techniques for Multimodal Detection of Bangla Fake News in Under-Resourced Contexts.**” [Paper Link](#) (Q1) 

Ongoing Research Projects

- FanSpeak: A Dataset on Toxicity in Sports-Related Online Discourse [*under review on [Information Processing & Management](#) , Q1 journal*]
- SarcFormer: A Transformer Based Framework for Sarcasm Analysis
- CoT-Driven Explainable Counter-Response Generation for Personalized Bullying Mitigation

Technologies

- **Programming Languages:** C++, C, Java, Python, SQL
- **Web Development:** HTML, CSS, Streamlit, FastAPI, Flask
- **Database:** MsSQL, MySQL
- **Deep Learning Framework:** TensorFlow, Keras, PyTorch
- **LLM Application Frameworks:** LangChain, LangGraph, LlamaIndex
- **LLMEvaluation Frameworks:** LangSmith, DeepEval
- **Vector Database:** ChromaDB, FAISS
- **Others:** Prompt Engineering, OpenCV, Machine Learning, Convolutional Neural Network

Reference

Dr. Shamim Akhter

Professor

Department of Computer Science and Engineering

Ahsanullah University of Science and Technology

Email: shamimakhter.cse@aust.edu 

Khan Md. Hasib

Assistant Professor

Department of Computer Science and Engineering

Ahsanullah University of Science and Technology

Email: khanmdhasib.aust@gmail.com 

Tashreef Muhammad

Lecturer

Department of Computer Science and Engineering

Southeast University

Email: tashreef.muhammad@seu.edu.bd 